Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9921669382128087	0.9164682291013913 – 1.0027768595041322

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9885282882882883	0.8586642621355661 – 1.079443083541639

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.209485633752933	1.1546833718414344 – 1.3779067180145486

Thin QR

512 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9936245424891716	0.8976748411295378 – 1.0586904929547531

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0182715183366788	0.9539091491294941 – 1.0669591519324717

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0408798988621997	0.9586351199098664 – 1.0546903947744524

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9951534420254735	0.961185853240413 – 1.0564999190545572

Thin QR

256 × 64 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9986932253431998	0.9374925164989188 – 1.0351285018554286

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9821038847664775	0.945438673068529 – 1

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.00001011807797	0.9388606238475425 – 1.0940139622787055

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9991894510013245	0.9028559547324492 – 1.0428752324799178

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9962362276819777	0.9468969486462286 – 1.0104691648198838

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0213066491832494	0.8922478299910207 – 1.1183833501174891

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9727626459143969	0.9375937593759376 – 0.9869077306733167

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.8936943272386311 – 1.0455315505683473

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	0.9991960784313726	95% INTERVAL	0.9549636566040207 – 1.0070588235294118

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9853099247581512	0.9565217391304348 – 1

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0474285714285714	0.9541984732824428 – 1.158669574700109

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9675904789281748	0.916102372139973 – 1.0378457059679767

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.000398406374502	0.9715976292854505 – 1.0930526315789473

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0001606038705533	0.9693697400669682 – 1.1191409331409332

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9864052621133509	0.4546756756756757 – 1.1396162361623616

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9985013499839345	0.7590126712952051 – 1.061062513641804

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9885894736842106	0.7937915742793792 – 1.2318772378516625